MODULE 2 BONE TISSUE

Bone Tissue and Microanatomy.

BONE HISTOLOGY

Bone and cartilage are the supportive connective tissues. This chapter covers cartilage, the structure of a bone from the whole organ down to the osteon, the four bone cells, and how bone forms and grows.

BY THE END

1. Describe the three types of cartilage and the cells and matrix that make up cartilage.
2. Classify bones by shape and identify the gross parts of a long bone.
3. Describe the microanatomy of compact bone, centred on the osteon, and contrast it with spongy bone.
4. Identify the four bone cell types and what each one does.
5. Outline how bones form and grow, using the growth plate and its zones in order.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Bone

drsrennie-stack.github.io/new-build-bio4-solano/bone-concept-videos.html

Cartilage

Cartilage is a firm, flexible connective tissue. It has no blood vessels of its own, so it heals slowly. That single fact explains most of what is clinically interesting about it.

THE CARTILAGE TISSUE

Component	What it is
Chondrocytes	The cartilage cells. Each one sits in a small space called a lacuna
Lacunae	The cavities in the matrix that house the chondrocytes
Matrix	A firm gel of ground substance with collagen, and in some types elastic fibers
Perichondrium	The connective tissue sheath wrapped around most cartilage

The three types of cartilage

FIBERS, TEXTURE, AND WHERE EACH IS FOUND

Type	Fibers and texture	Where it is found
Hyaline cartilage	Fine collagen, glassy and smooth. The most common type	Ends of long bones, the nose, the airways, the costal cartilages
Elastic cartilage	Many elastic fibers, flexible and springy	The external ear and the epiglottis
Fibrocartilage	Thick bundles of collagen, tough and shock-absorbing	The intervertebral discs, the knee menisci, the pubic symphysis

How cartilage grows

TWO DIRECTIONS OF GROWTH

Type of growth	What happens	Where it starts
Appositional	New cartilage is added at the outer surface	From the perichondrium
Interstitial	The cartilage expands from within	Chondrocytes divide inside the matrix

Bone Classification and the Long Bone

Bones by shape

FIVE SHAPES, WITH EXAMPLES

Shape	Description	Examples
Long	Longer than they are wide, a shaft with two ends	Femur, humerus
Short	Roughly cube-shaped	Carpals, tarsals
Flat	Thin and often curved	Sternum, ribs, most skull bones
Irregular	Complex shapes that fit no other group	Vertebrae, hip bones
Sesamoid	Small bones formed within a tendon	Patella

Gross anatomy of a long bone

FROM THE SHAFT OUTWARD

Structure	What it is
Diaphysis	The shaft, a tube of compact bone around the medullary cavity
Epiphysis	Each expanded end. Spongy bone under a thin compact shell
Metaphysis	The region between diaphysis and epiphysis. Holds the growth plate or its remnant
Medullary cavity	The hollow core of the diaphysis, holds marrow
Articular cartilage	Hyaline cartilage capping the epiphysis where it meets another bone
Periosteum	The connective tissue membrane covering the outer bone surface
Perforating (Sharpey) fibers	Bundles of collagen that anchor the periosteum into the underlying bone
Nutrient foramen	The opening in the shaft where the nutrient artery enters the bone
Endosteum	The thin membrane lining the internal bone surfaces
Epiphyseal line	The remnant of the growth plate after growth in length ends
Red marrow	Fills the spongy bone of the epiphyses. The site of blood cell formation
Yellow marrow	Fat stored in the medullary cavity of the adult shaft

Bone Microanatomy

Compact bone

THE OSTEON AND WHAT SURROUNDS IT

Structure	What it is
Osteon	The Haversian system, the structural unit of compact bone. A set of tubes around a canal
Central canal	The Haversian canal at the core of the osteon. Carries blood vessels and nerves
Perforating canal	The Volkmann canal. Runs crosswise and links central canals
Lamellae	The concentric rings of bony matrix around the central canal
Lacunae	Small cavities between the lamellae. Each holds one osteocyte
Canaliculi	Tiny channels that connect lacunae and let osteocytes exchange materials
Interstitial lamellae	Leftover fragments of old osteons, filling the gaps between intact osteons
Circumferential lamellae	Rings of bone that wrap the entire outer and inner surface of the shaft

Spongy bone

WHAT REPLACES THE OSTEON

Structure	What it is
Trabeculae	The open lattice of bony struts that forms spongy bone, aligned along lines of stress
No osteons	Spongy bone has no osteons. Its spaces hold red marrow

HOLD ONTO THIS

Compact bone solves the problem of strength with the osteon. Spongy bone solves the problem of weight with the trabecula. Same tissue, two answers, and you can tell them apart on a slide by looking for a central canal.

The four bone cells

CELL, JOB, AND WHERE IT COMES FROM

Cell	What it does	Origin or fate
Osteogenic cell	Divides to produce new osteoblasts	The stem cell of bone tissue
Osteoblast	Builds and secretes new bone matrix	Becomes an osteocyte once surrounded by matrix
Osteocyte	Maintains the bone matrix day to day	A mature osteoblast, sitting in a lacuna
Osteoclast	Breaks down and resorbs bone matrix	A large multinucleated cell from a blood-cell line, unlike the other three

HOLD ONTO THIS

Three of the four are one lineage: osteogenic becomes osteoblast becomes osteocyte. The osteoclast is the outsider, from a blood-cell line, and it is the only one that takes bone away.

Bone Growth and Formation

This section follows the structures and the sequence: the two ways bone forms, the two ways it grows, and the ordered zones of the growth plate. The chemistry belongs to physiology and is not covered here.

Ossification, the two routes that form bone

WHAT EACH ROUTE BUILDS

Route	What it builds	In brief
Intramembranous	The flat bones of the skull	Bone forms directly within a connective tissue membrane
Endochondral	Most bones of the body	Bone replaces a hyaline cartilage model

The two ways a long bone grows

The **epiphyseal plate** is the disc of hyaline cartilage where lengthening happens. When growth ends, the plate closes and leaves the **epiphyseal line**.

DIRECTION AND LOCATION

Type of growth	Direction	Where it happens
Interstitial growth	The bone grows longer	At the epiphyseal plate
Appositional growth	The bone grows wider	At the outer surface, beneath the periosteum

Zones of the growth plate

Five zones in order, from the epiphysis at the top to the diaphysis at the bottom.

1. **Zone of resting cartilage.** Anchors the growth plate to the epiphysis.
2. **Zone of proliferation.** Chondrocytes divide and stack into columns.
3. **Zone of hypertrophy.** The chondrocytes enlarge.
4. **Zone of calcification.** The cartilage matrix calcifies.
5. **Zone of ossification.** Cartilage is replaced by bone next to the diaphysis.

Classification: the supportive connective tissues

Bone histology fits inside one family. This map ties the chapter together: cartilage and the two kinds of bone, each with its structural unit.

THE FAMILY, WITH EACH STRUCTURAL UNIT

Tissue	Its members or its unit
Cartilage	Hyaline, elastic, fibrocartilage
Compact bone	Built from osteons: central canal, lamellae, lacunae, canaliculi
Spongy bone	Built from trabeculae, with red marrow in the spaces

BONE STRUCTURE SHOWS UP IN EVERY X-RAY

The growth plate is cartilage, so on a child's X-ray it looks like a gap. A fracture that runs through the plate can slow or stop a bone from growing, which is why pediatric fractures are described by how they cross it. The epiphyseal line is the closed plate, and its presence tells a clinician the skeleton is mature. Compact bone is built from osteons aligned with the lines of stress, so a bone is strongest along its long axis. That is why a twisting force, off that axis, is the one that tends to break it.